The study of **joint attention** benefits by taking a novel **multisensory approach**

Joint attention beyond vision

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1. INTRODUCTION

- **Joint attention** refers to two or more individuals coordinate their attention to a third object or event.
- Operationalisations rely on gaze following, gaze alternation and pointing.
- It is considered fundamental for *language learning*, for *understanding others* and for *joint actions*.[1]

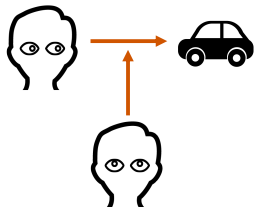
What is the role of non-visual senses for this achievement?

2. OBJECTIVES

- Outline a framework that takes into account the complexity of multisensory social attention.
- Identify the unique contributions of non-visual senses.
- Show why a multisensory approach is needed to study new facets of coordinated attention.

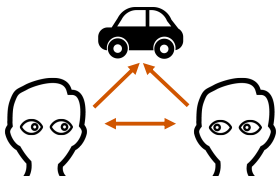
3. VISUAL JOINT ATTENTION

DYADIC FOLLOWING OF ATTENTION



- Unilateral response of one individual
- Relies on perceptual skills[1]
- E.g. Gaze and face processing

TRIADIC GENUINE JOINT ATTENTION

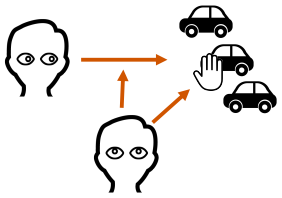


- Reciprocal interaction between individuals
- Requires social understanding[1]
- E.g. gaze alternation between object and partner

4. INSTRUMENTAL ROLE: WHY NON-VISUAL SENSES ARE *USEFUL*

Non-visual cues act as scaffolders for gaze and pointing gestures to establish, maintain joint attention to **visible** targets

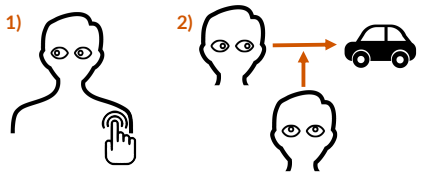
REDUCTION OF SPATIAL UNCERTAINTY



Redundant information delivered across modalities:

- Increased accuracy and speed in following spatial cues[2]

ATTRACTING THE OTHER'S GAZE



Complementary cues with different information:

- Attention-getter in one modality (1) to then direct attention more effectively in another modality (2)

5. SOCIAL ROLE: WHY NON-VISUAL SENSES ARE *NECESSARY*

COORDINATING ATTENTION on NON-VISUAL TARGETS

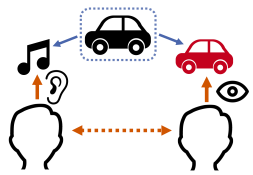
Ostensive use of visual cues:

- Relies on cognitive strategies to infer that the target is non-visual
- Presupposes understanding that sounds / smells can be shared.

ATTENDING to WHOLE OBJECTS vs. SENSIBLE FEATURES

Coordinating on amodal features

- Strategic use of temporal / spatial synchrony to bind sensory stimuli to the same object.
- Enhance association of object and word[3]



Coordinating on objects we each experience through different modalities:

- Each subject must process information about each other's modal access relative to the target

6. PRACTICAL AND THEORETICAL IMPACT

SOCIAL SELECTIVITY

- The *necessary* role of non-visual senses in joint attention highlights the **social selectivity of attention.**
- Selective processing of the target of coordinated attention can be narrower / broader than the object of individual sensory attention.

OUTSTANDING QUESTIONS

- Does the strategic use of multisensory interactions underlie heightened processing of jointly attended targets?
- Hypothesis:

Crossmodal binding is enhanced in condition 1 over 2 and 3:

1. **Joint** attention + temporally **aligned** audiovisual stimuli
2. **Non-joint** attention + temporally **aligned** audiovisual stimuli
3. **Joint** attention + temporally **misaligned** audiovisual stimuli

ARTIFICIAL AGENTS

- **Challenge**: Artificial agents must display the social understanding manifested in human joint attention, and meet human's behavioural expectations.[4]
- The coordinated exercise of sensory modalities can assist in addressing this challenge.

[1] Mundy, P. (2018). A review of joint attention and social-cognitive brain systems in typical development and autism spectrum disorder. *Eur J Neurosci*, *47*(6).
[2] Ernst, M. O., & Bülthoff, H. H. (2004). Merging the senses into a robust percept. *Trends Cogn Sci*, 8(4), 162–169.
[3] Gogate, L., & Hollich, G. (2016). Early verb-action and noun-object mapping across sensory modalities. *Dev Neuropsychol*, 41(5–8), 293–307.
[4] Kaplan, F., & Hafner, V. V. (2006). The challenges of joint attention. *Interact Stud*, 7(2), 135–169.

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